and

$$\partial_{\theta_i}\partial_{\theta_j}\mathsf{E}_\beta(\theta_1,\dots,\theta_n) = \frac{1}{n^2}\cdot\begin{cases} g(\theta_i-\theta_j), & i\neq j \\ -\sum\limits_{m\in[n]\setminus\{i\}} g(\theta_i-\theta_m), & i=j\,, \end{cases}$$

where we set $g(x) := (\cos(x)-\beta\sin^2(x))e^{\beta\cos(x)}$. Plugging this expression back into (B.1) and simplifying, we obtain

$$\sum_{i\in\mathcal{S}}\sum_{j\in\mathcal{S}^c} g(\theta_i-\theta_j) \geqslant 0\,. \tag{B.2}$$

Let us now define τ_β^* be the unique solution on $[0,\frac{\pi}{2})$ of the equation

$$\beta\sin^2(\tau) = \cos(\tau)\,.$$

Note that τ_β^* is a monotonically decreasing function of β, and in fact

$$\tau_\beta^* = \frac{1+o(1)}{\sqrt{\beta}}$$

as $\beta\to+\infty$. The importance of τ_β^* is in implying the following property of the function g: for any $\tau\notin[-\tau_\beta^*,\tau_\beta^*]$, we must have that $g(\tau)<0$ (see Figure 6). We arrive at the following conclusion: it must be that for any proper subset $\mathcal{S}\subset[n]$ there exists, by virtue of (B.2), some index $j\in\mathcal{S}^c$ such that

$$\inf_{k\in\mathcal{S}}|\theta_j-\theta_k| < \tau_\beta^*.$$

So now let us start with $\mathcal{S}=\{1\}$ and grow $\mathcal{S}$ inductively by adding those points θ_j at distance $<\tau_\beta^*$ from $\{\theta_k\colon k\in\mathcal{S}\}$ at each induction step. If β is large enough so that

$$(n-1)\tau_\beta^* < \frac{\pi}{2},$$

then in the process of adding points we have travelled a total arc-length $<\pi/2$ on each side of x_1. Thus it must be that the collection of points $\theta_1,\dots,\theta_n$ is strictly contained inside a half-circle of angular width $<\pi$. By Lemma 6.4 we know that there can be no critical points of E_β that are strictly inside some half-circle, unless that critical point is trivial: $\theta_1=\cdots=\theta_n$. This completes the proof when $d=2$.

We can show that the same conclusion holds for any dimension $d\geqslant 2$. The proof follows by arguing just as above, making instead use of the following generalization of (B.2): given a collection $x_1,\dots,x_n\in\mathbb{S}^{d-1}$ at which the Hessian of E_β is non-positive, we must have for any subset $\mathcal{S}\subset[n]$ that

$$\sum_{i\in\mathcal{S}}\sum_{j\in\mathcal{S}^c} g(\theta_{ij}) \geqslant 0, \tag{B.3}$$

where $g(\zeta) = e^{\beta\cos(\zeta)}((d-1)\cos(\zeta)-\beta\sin^2(\zeta))$ and $\theta_{ij}\in[0,\pi]$ is the geodesic distance between x_i and x_j, namely $\cos(\theta_{ij})=\langle x_i,x_j\rangle$. We now show (B.3). By repeating the argument in Step 2 of the proof of Lemma A.2 , we see that for any skew-symmetric matrix B we must have

$$\sum_{i\in\mathcal{S}}\sum_{j\in\mathcal{S}^c} e^{\beta\langle x_i,x_j\rangle}\Big(\beta\langle Bx_i,x_j\rangle^2+\langle B^2x_i,x_j\rangle\Big) \leqslant 0. \tag{B.4}$$

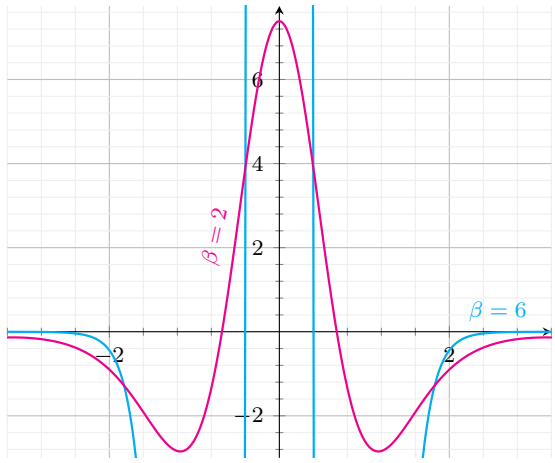

Figure 6. The function $\tau \mapsto g(\tau)$ for two values of β.

Now we take B to be random by generating $B_{ij} \overset{\text{i.i.d.}}{\sim} P$, $i < j$ and P being any zero-mean, unit-variance distribution. We set $B_{ji} = -B_{ij}$ and $B_{ii} = 0$. Then it is easy to check that

$$\mathbb{E}[B^2] = -(d-1)I_d$$

and

$$\mathbb{E}[\langle Bx_i, x_j\rangle^2] = 1 - \langle x_i, x_j\rangle^2 = \sin^2(\theta_{ij}).$$

Thus, taking the expectation over all such B in (B.4) yields (B.3). Mirroring the proof for $d = 2$, we define τ_β^* to be the unique solution on $[0, \frac{\pi}{2})$ of the equation $\beta \sin^2(\tau) = (d-1)\cos(\tau)$. We note that

$$\tau_\beta^* = \sqrt{\frac{(d-1) + o(1)}{\beta}}$$

for $\beta \to +\infty$. Repeating verbatim the argument for the case $d = 2$, we deduce the convergence to a single cluster whenever $\beta \gtrsim (d-1)n^2$. □

Remark B.1. *We comment on the extension of the above proof to the dynamics* (SA)*. We recall that* (SA) *is a gradient flow, but for a different metric—see Section 3.4—and we show that the saddle point property is preserved across metrics. Our proof is an adaptation of a classical argument: the Hessian of a function at a critical point is a notion which does not depend on the choice of Riemannian metric.*

Let $x = (x_1, \dots, x_n) \in (\mathbb{S}^{d-1})^n$ be a critical point of E_β (this does not depend on the metric) such that not all x_i are equal to each other. Recall that for $f : (\mathbb{S}^{d-1})^n \to \mathbb{R}$, for any metric on $(\mathbb{S}^{d-1})^n$ (with associated Christoffel symbols Γ_{ij}^k) and any associated orthonormal basis $y_1, \dots, y_{(d-1)n}$, the Hessian of f reads

$$\mathrm{Hess}(f) = \left(\frac{\partial^2 f}{\partial y_i \partial y_j} - \Gamma_{ij}^k \frac{\partial f}{\partial y_k}\right) dy_i \otimes dy_j. \tag{B.5}$$

Since we are evaluating the Hessian at a critical point x of E_β, the term carrying the Christoffel symbols Γ_{ij}^k vanishes. In the above argument, we saw that $\mathrm{Hess}\,\mathsf{E}_\beta$ *evaluated at x, and written in an orthonormal basis for the canonical metric g on*

$(\mathbb{S}^{d-1})^n$, is not negative semi-definite. We denote this matrix by M_g; we know that there exists $v \in \mathrm{T}_x(\mathbb{S}^{d-1})^n$ such that $g(v,v) = 1$ and $v^\top M_g v > 0$. Let $\tilde{g}$ be another metric on $(\mathbb{S}^{d-1})^n$; we denote by $M_{\tilde{g}}$ the Hessian evaluated at x, and written in an orthonormal basis for $\tilde{g}$. Let $c : \mathbb{R}_{\geqslant 0} \to (\mathbb{S}^{d-1})^n$ be such that $c(0) = x$ and $\dot{c}(0) = v$. Since x is a critical point (for both metrics), a Taylor expansion to second order in the two orthonormal bases yields

$$\mathsf{E}_\beta(c(t)) = \mathsf{E}_\beta(c(0)) + \frac{1}{2}t^2 v^\top M_g v + O(t^3)$$

as well as

$$\mathsf{E}_\beta(c(t)) = \mathsf{E}_\beta(c(0)) + \frac{1}{2}t^2 \|v\|_{\tilde{g}}^{-2} v^\top M_{\tilde{g}} v + O(t^3)$$

thanks to (B.5). Hence $v^\top M_{\tilde{g}} v > 0$. Specializing to $\tilde{g}$ being the metric of Section 3.4, with respect to which (SA) is a gradient flow for E_β, we conclude for (SA).

Appendix C. Proof of Theorem 4.3

Proof of Theorem 4.3. We leverage the gradient flow structure and follow the same strategy as in the proof of Theorem 5.1 presented above. For simplicity we write the argument for (USA) and explain the extension to the case of (SA) in Remark C.1.

Consider

$$\mathsf{E}_\beta(x_1, \dots, x_n) = \frac{1}{2\beta} \sum_{i=1}^n \sum_{j=1}^n \left(e^{\beta \langle x_i, x_j \rangle} - 1 \right).$$

Note that this is only a slight deviation from the energy studied in Section 3.4: we solely subtracted a constant. Consequently the dynamics (USA) are also a gradient flow for this energy. The main interest of considering this modified energy is the observation that

$$\mathsf{E}_\beta(x_1, \dots, x_n) = \mathsf{E}_0(x_1, \dots, x_n) + \beta\, \mathsf{R}_\beta(x_1, \dots, x_n),$$

where R_β is smooth. Hence R_β has a bounded Hessian on $(\mathbb{S}^{d-1})^n$ uniformly with respect to β, and

$$\nabla \mathsf{E}_\beta = \nabla \mathsf{E}_0 + O(\beta), \qquad \operatorname{Hess} \mathsf{E}_\beta = \operatorname{Hess} \mathsf{E}_0 + O(\beta). \tag{C.1}$$

Observe that in the proof of Theorem 4.1, we actually showed that there exists $c > 0$ such that at any critical point $(x_1, \dots, x_n)$ of E_0 for which $x_i \neq x_j$ whenever $i \neq j$, at least one of the eigenvalues of the Hessian of E_0, λ say, satisfies $\lambda \geqslant c$. Indeed, in (A.2) the proof actually shows the existence of some $\mathcal{S} \subset [n]$ such that

$$\sum_{i \in \mathcal{S}} \sum_{j \in \mathcal{S}^c} \langle x_i, x_j \rangle \leqslant -1.$$

Then, (A.3), together with (A.4) for instance, yield

$$\mathsf{E}_0''(0) \geqslant \frac{2(d-1)}{dn} =: c \tag{C.2}$$

for one of the B_j.

Now suppose that there exists a positive sequence $\beta_k \to 0$ as well as $X_k \in (\mathbb{S}^{d-1})^n$ such that X_k is a critical point of E_{β_k} and all of the eigenvalues of $\operatorname{Hess} \mathsf{E}_{\beta_k}(X_k)$ are non-positive. Then by virtue of the continuity properties of E_β with respect to β in (C.1), we find that, up to extracting a subsequence, there is some limit point $\overline{X} = (x_1, \dots, x_n) \in (\mathbb{S}^{d-1})^n$ of X_k which is a critical point of E_0, and such that all of